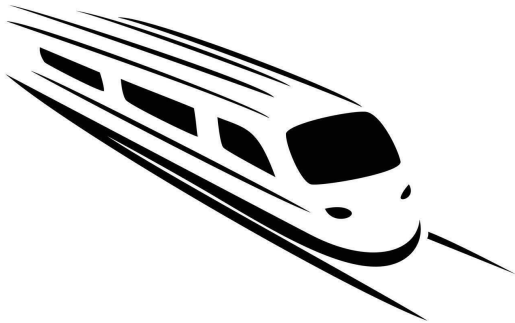


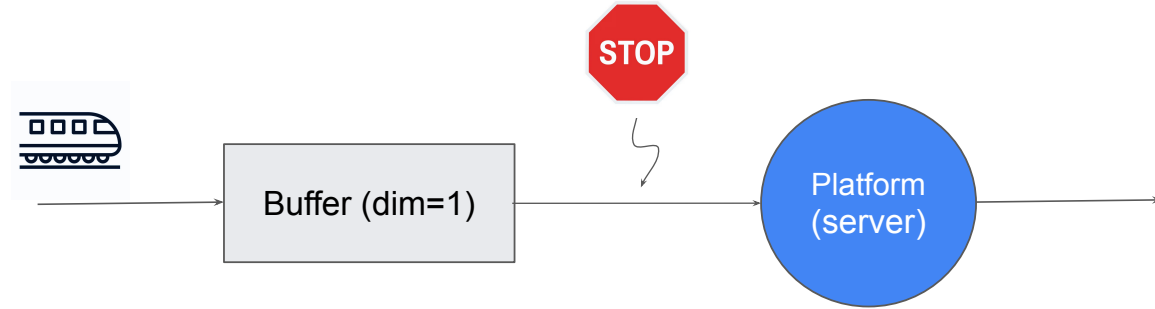
D.E.S. model of a train station

Final report



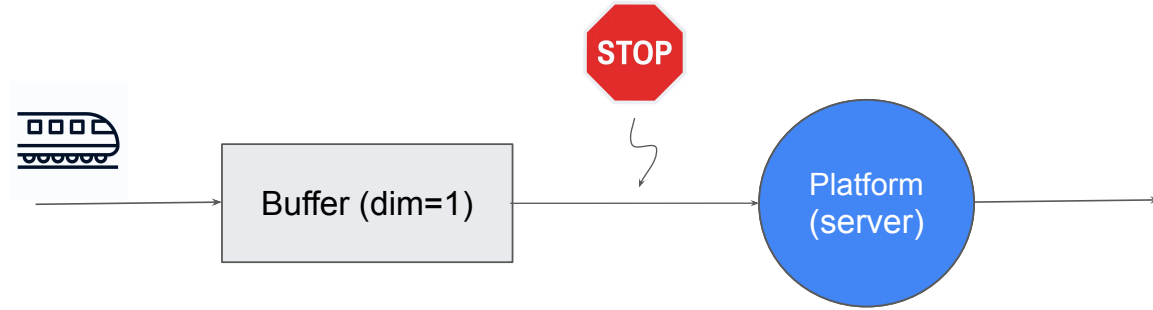
Valentino Basili

Situation



- A single-dimensional buffer that can hold a waiting train.
- A platform where the train stops (passengers can get off the train), pick up passengers, and depart.
- A stop signal that lights up in the event of a fault on the line and blocks access to the platform.

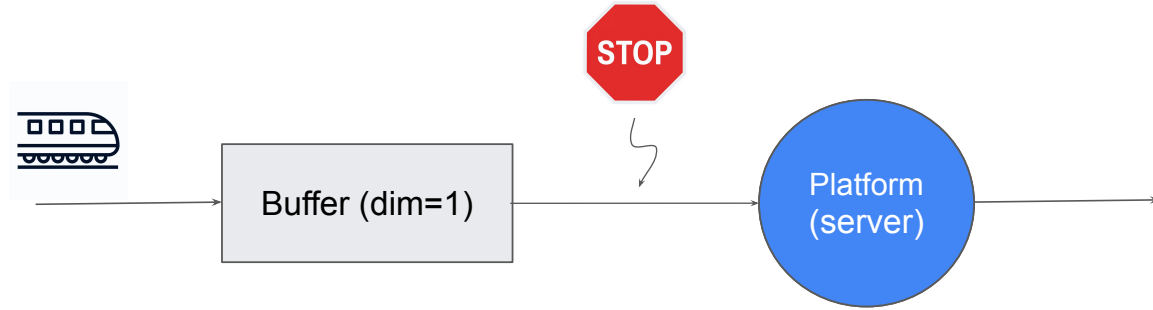
About



The occurrence of a fault instantly interrupts all the operations, "freezing" the train in its current phase and blocking access from the queue. During the recovery time, the only permitted network event (other than the repair itself) is the queuing of a new arrival, if the buffer is empty.

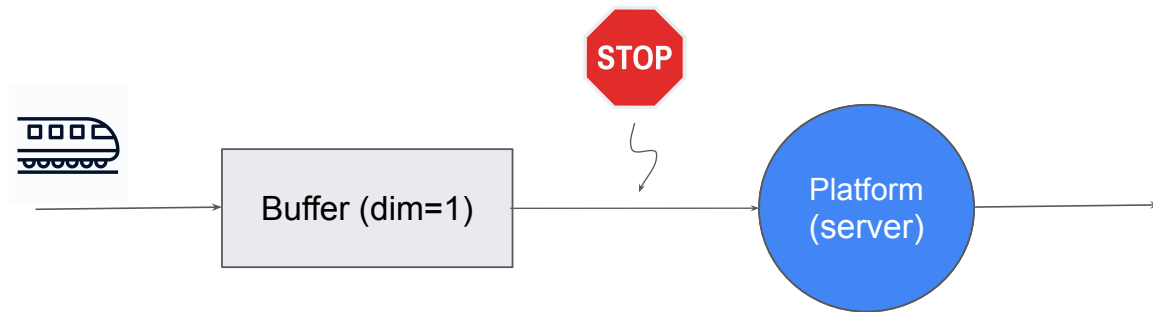
Assume the train station is initially empty.

Events

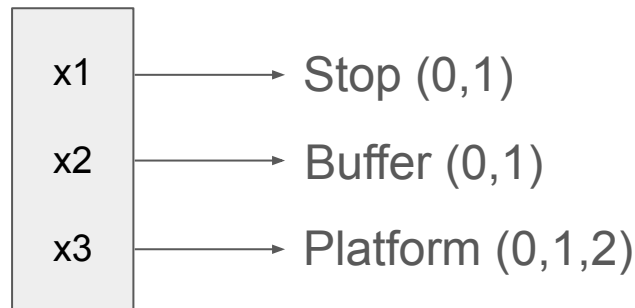


- a : new train arrives.
- c : the passengers got off and new passengers can board.
- d : train departs.
- g : fault occurs.
- r : repairs are complete.

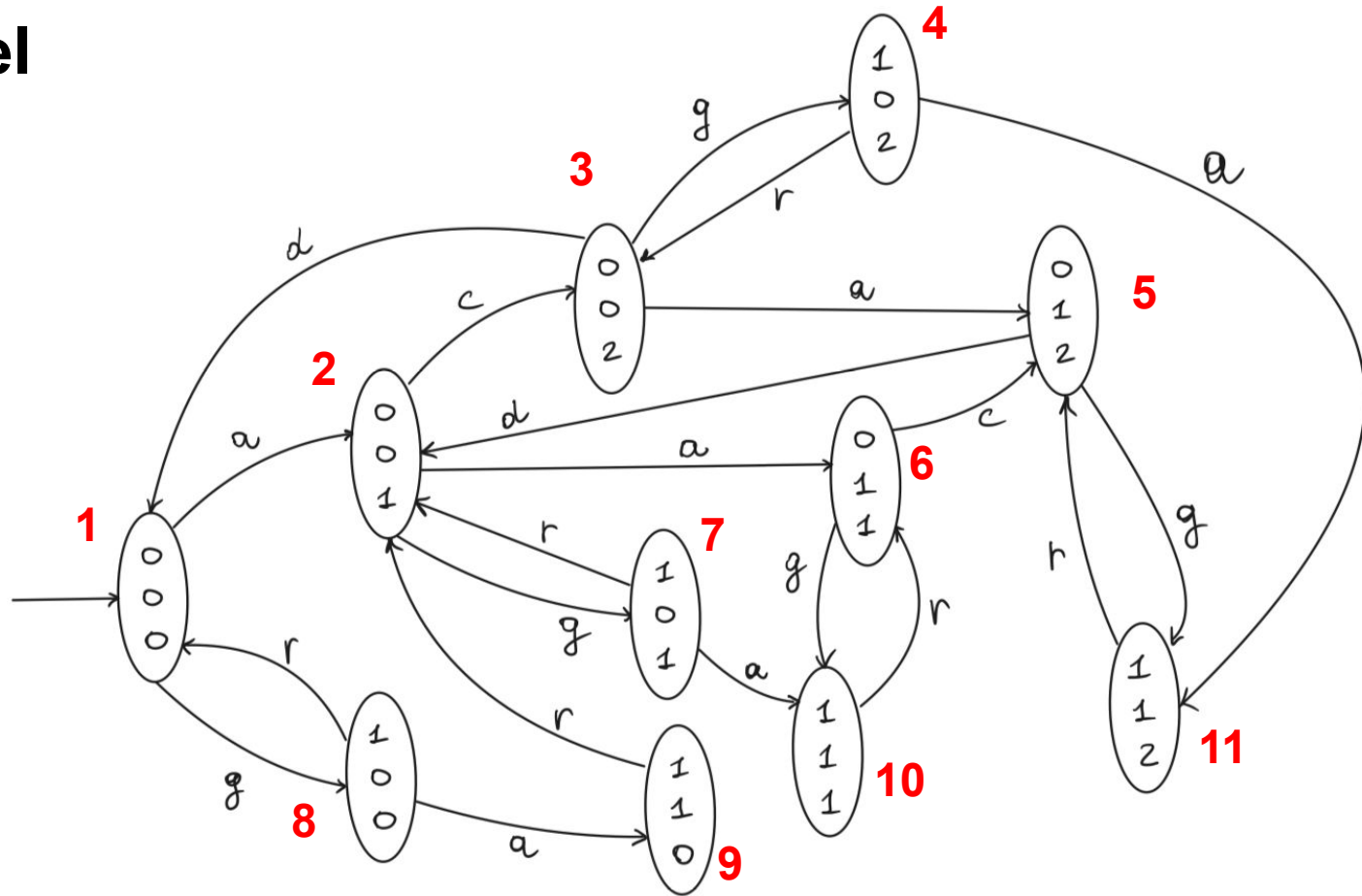
States



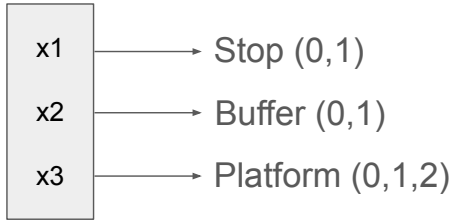
- Stop signal can be 0 (free) or 1 (failure).
- Buffer can be 0 (empty) or 1 (full).
- Platform can be 0 (empty),
1 (passengers getting off the train),
2 (train ready).



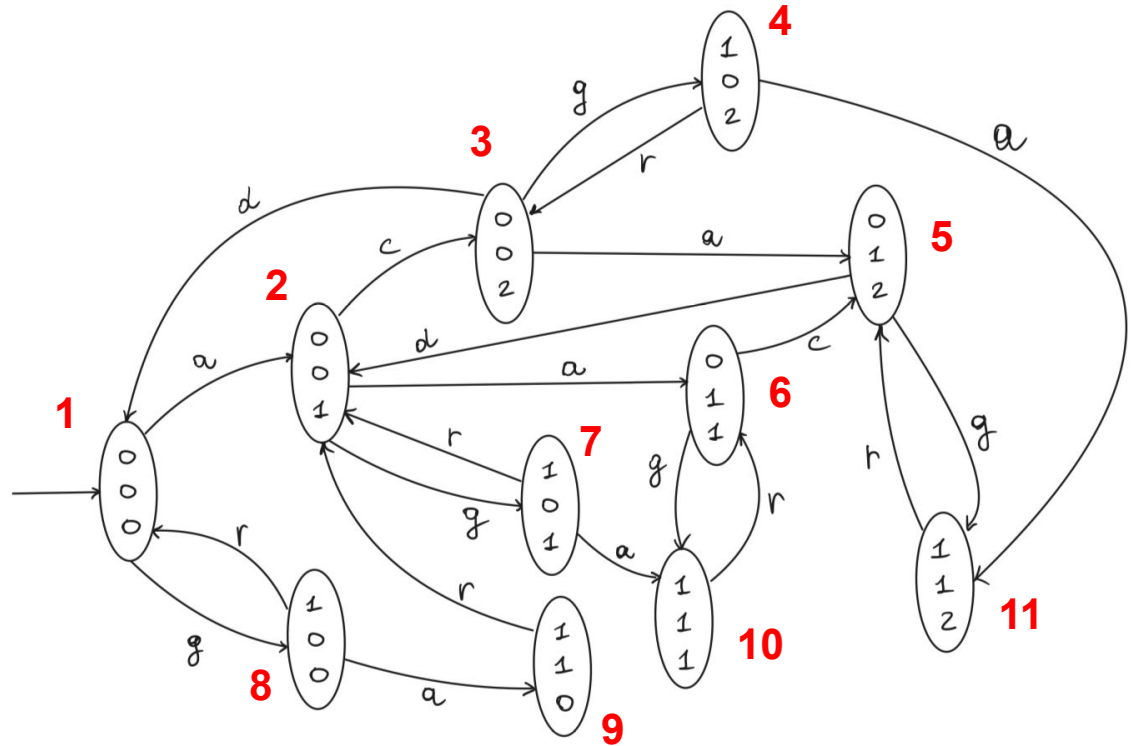
Model



Overview



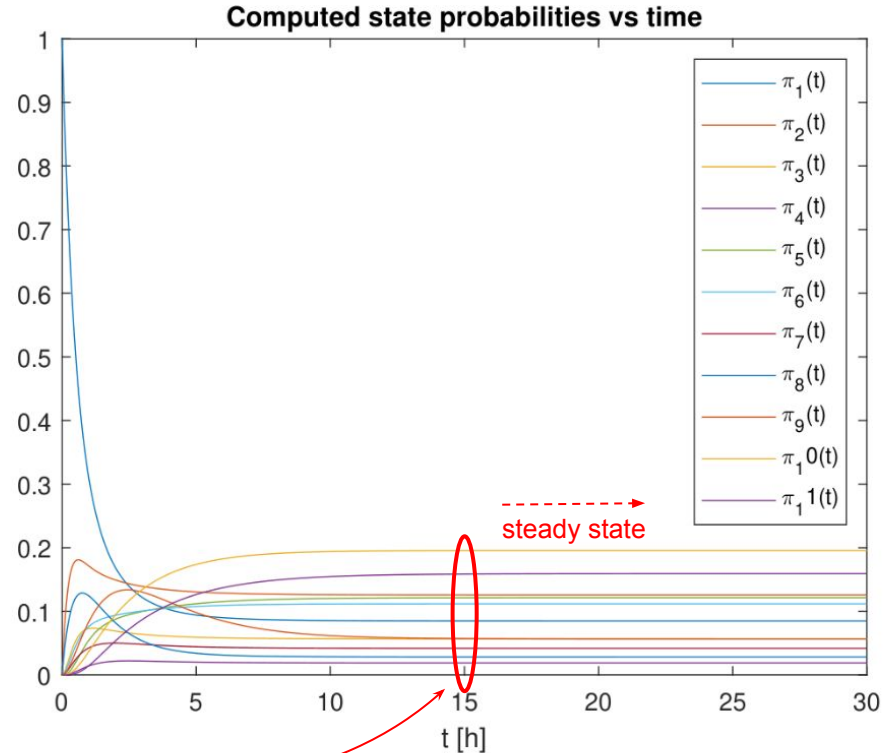
- a : new train arrives.
- c : the passengers got off and new passengers can board.
- d : train departs.
- g : fault occurs.
- r : repairs are complete.



Poisson clock structure

Since our stochastic state automaton with Poisson clock structure is stochastic equivalent to a finite irreducible CTHMC, we can **compute** limit state probabilities.

To do this we calculate $\pi(0)e^{Qt}$ until time is equal to 30 [h]: after all this time the system will in practice already be in steady state, as we can see from the graph:



Poisson clock structure

The values of $\pi(t)$ that we obtain at $t=30$ [h], since the system is at steady state after $t \approx 15$ [h] are our **limit state probabilities**. In particular, the values are the following:

Limit state probabilities computed analytically:

```
pi = [0.084787    0.12561    0.056525    0.018842    0.12142    0.11165    0.04187    0.028262    0.056525    0.19539    0.15911]
```

$$\pi(t) = [\pi(1) \quad \pi(2) \quad \pi(3) \quad \pi(4) \quad \pi(5) \quad \pi(6) \quad \pi(7) \quad \pi(8) \quad \pi(9) \quad \pi(10) \quad \pi(11)]$$

Poisson clock structure

To test whether we are simulating correctly, we **estimate** the probabilities of the limit state and check whether they are the same as those we calculated.

To do this, we run N independent simulations (all starting from the same initial state) and, for each time instant, calculate the estimated probability of a state by counting the fraction of those N simulations that are in that state at that exact moment.

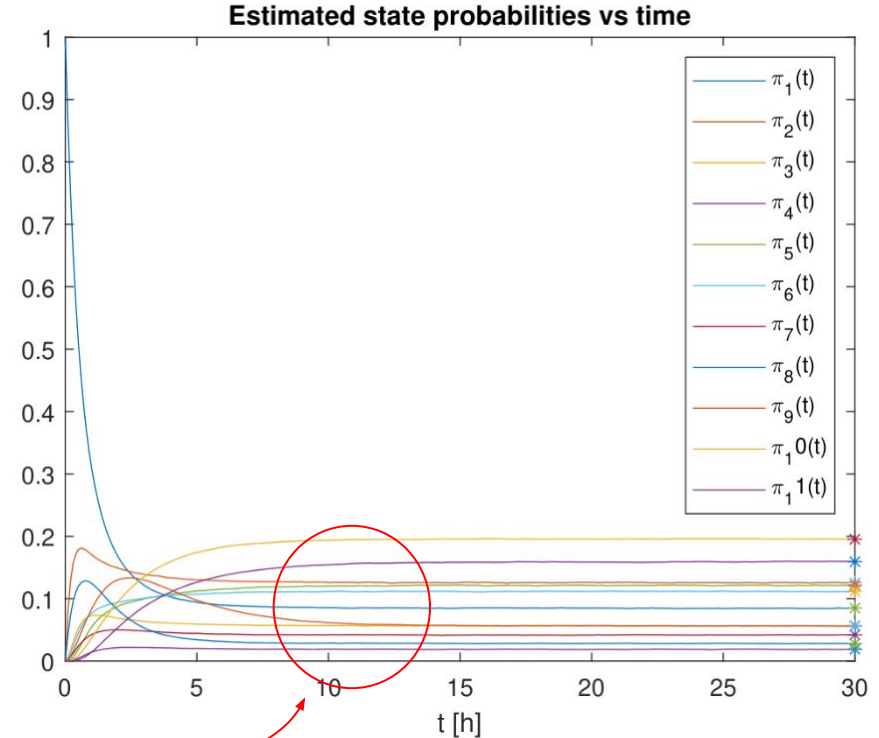
Since the estimation error is inversely proportional to the square root of the number of simulations N , to obtain an estimation error less than 10^{-3} we need to run 10^6 independent simulations.

Poisson clock structure

If we plot the values of the estimated probabilities we get the following graph:

As we can see again the steady state is reached after $t \approx 15$ [h] so we can take the values of the estimated probabilities at $t = 30$ [h] and compare them with the computed values.

If we zoom in we can see very small oscillations:



Poisson clock structure

Computed limit state probabilities:

Limit state probabilities computed analytically:

```
pi = [0.084787    0.12561    0.056525    0.018842    0.12142    0.11165    0.04187    0.028262    0.056525    0.19539    0.15911]
```

Estimated limit state probabilities:

```
px_est =
```

```
0.0849    0.1253    0.0565    0.0190    0.1214    0.1119    0.0421    0.0281    0.0562    0.1952    0.1593
```

The calculated and estimated probabilities of the limit state match almost perfectly (estimation error less than 10^{-3}) successfully verifying the correctness of the simulation logic. This allows us to adopt arbitrary realistic distributions.

Evidence of Law of Large Numbers (Poisson case)

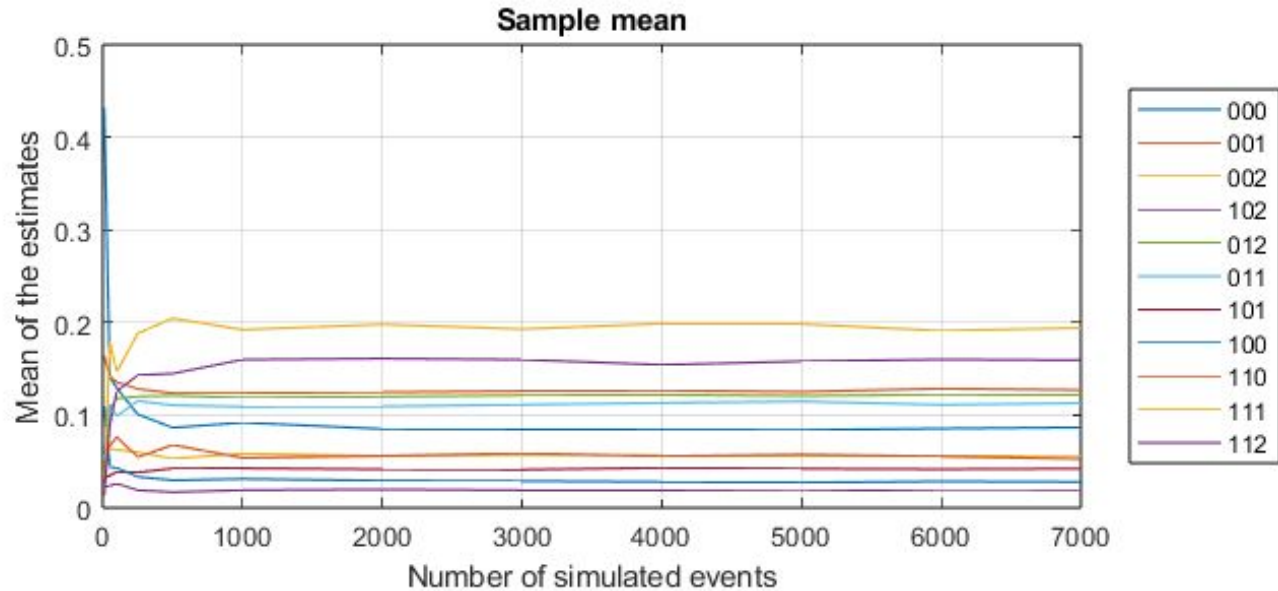
To give evidence of the Law of Large Numbers we perform $M=100$ independent replications of the system and progressively increase the number of events before the simulation ends.

The probability of each state is now estimated by dividing the total time spent in a specific state by the total simulation time.

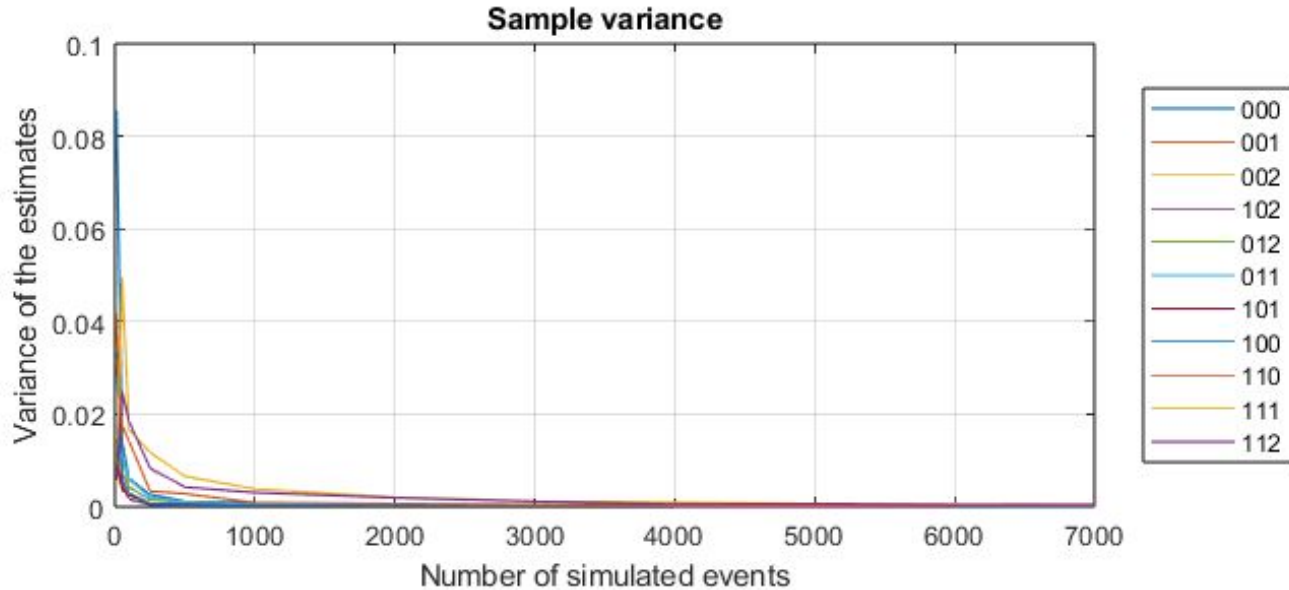
In the following plots we will see that as k (number of events before simulation ends) increases, the sample variance across the 100 replications significantly drops, while the sample mean converges to the exact limit values.

The number of events k assume several values from 10 to 7000.

Evidence of Law of Large Numbers (Poisson case)



Evidence of Law of Large Numbers (Poisson case)



Realistic clock structure

Since we are sure that our simulation logic is correct we can finally try to adopt realistic probability density functions for the clock structures of the events.

In particular (unit of measurement is [hours]):

- $a \sim \text{Exp}(1)$
- $c \sim U(0.2, 0.4)$
- $d = \{1.5, 1.5, 1.5, 1.5, \dots\}$
- $g \sim \text{Exp}(0.5)$
- $r \sim U(0.5, 1.5)$

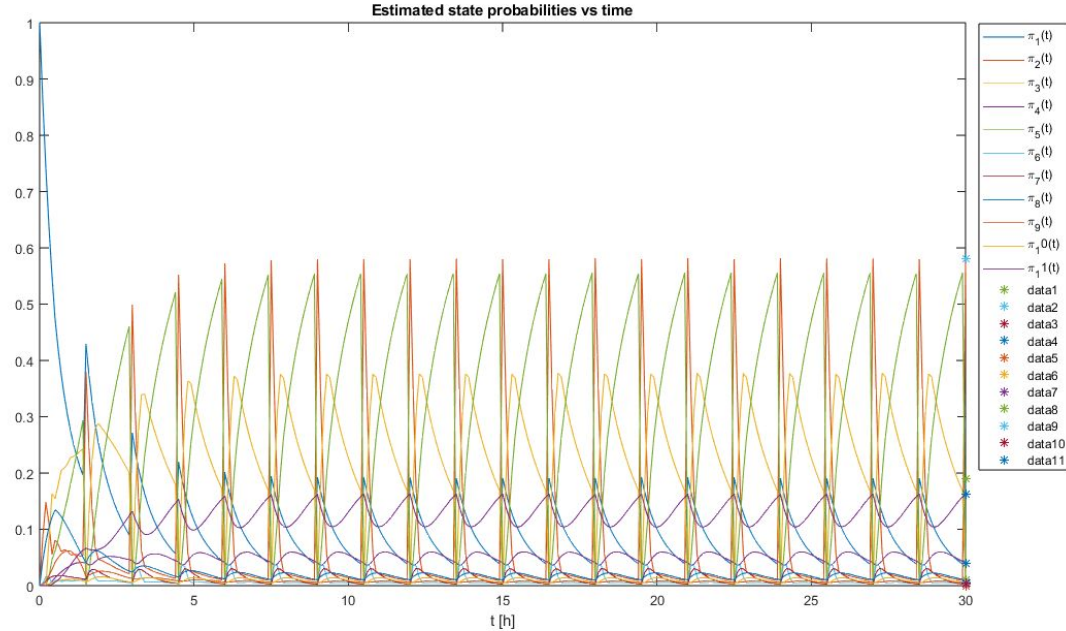
We run again $N=10^6$ independent simulations and plot the values of the estimated probabilities.

Realistic clock structure

The introduction of real clock structures generates oscillations.

The system does not converge to constant state probabilities but enters a periodic regime.

The initial transient is exhausted after about 5 hours, after which the oscillations stabilize.



Evidence of Law of Large Numbers (Realistic case)

However, also in this case we can give evidence of the law of large numbers, using the same script used previously and following the exact same procedure.

The only thing we change are the clock structures and its parameters.

We perform $M=100$ independent replications of the system and progressively increase the number of events before the simulation ends, as we did before. Again, k takes several values from 10 to 7000.

Evidence of Law of Large Numbers (Realistic case)

As we expected, the graphs follow the same trends we saw previously, giving the evidence of the law of large numbers.

